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Target audience: EWG

Allow multiple init-statements

Proposal: Wherever the language allows or requires an *init-statement*, to allow multiple such statements.

Motivation: An *init-statement* is required in `for` statements and allowed in range-based-`for`, `if` and `switch` statements. An *init-statement* is commonly used to declare, and optionally initialize, one or more variables. However, because at most one such statement is currently allowed, two or more variables can only be declared if they happen to be of the same type. This restriction seems to be an accident of syntax rather than a design intention. No syntactic ambiguity would arise from allowing multiple *init-statements* wherever the language currently allows or requires one *init-statement*.

Example of use:

```
for (int i = 0, j = 0; i < m; i++)           //OK in C++20
for (int k = 0; double s = 0.; k < n; k++)    //error in C++20, allowed under this proposal
```

Effect on existing code: None. Code using the proposed feature is ill-formed under C++20 and earlier standards.

Wording: (edits to N4910)

8 Statements	[stmt.stmt]
8.1. Preamble	[stmt.pre]
1	...
	<i>init-statement-seq</i> ;
	<i>init-statement-seq</i> _{opt} <i>init-statement</i>
	...
2	A <i>substatement</i> of a <i>statement</i> is one of the following:
(2.1)	— for a <i>labeled-statement</i> , its <i>statement</i> ,
(2.2)	— for a <i>compound-statement</i> , any <i>statement</i> of its <i>statement-seq</i> ,
(2.3)	— for a <i>selection-statement</i> , any of its <i>statements</i> (but not <i>an init-statement in its init-statement-seq</i>), or
(2.4)	— for an <i>iteration-statement</i> , its <i>statement</i> (but not <i>an init-statement in its init-statement-seq</i>).
3	A <i>statement</i> S1 <i>encloses</i> a <i>statement</i> S2 if
(3.1)	— S2 is a <i>substatement</i> of S1,
(3.2)	— S1 is a <i>selection-statement</i> or <i>iteration-statement</i> and S2 is <i>an the init-statement in the init-statement-seq</i> of S1,
(3.3)	— S1 is a <i>try-block</i> and S2 is its <i>compound-statement</i> or any of the <i>compound-statements</i> of its <i>handlers</i> , or
(3.4)	— S1 <i>encloses</i> a <i>statement</i> S3 and S3 <i>encloses</i> S2.

8.5. Selection statements	[stmt.select]
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{ Replace all occurrences of *init-statement* with *init-statement-seq*, except in Note 1 of 8.5.1. }

8.6. Iteration statements	[stmt.iter]
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{ Replace all occurrences of *init-statement* with *init-statement-seq*, except in Note 1 of 8.6.1. }

A.6 Statements

[gram.stmt]

...

init-statement-seq;

*init-statement-seq*_{opt} *init-statement*

Explanation: The proposed syntax will have the effect that:

- (i) all code within the main parentheses of a for statement, up to and including the penultimate semicolon, consists of init-statements;
- (ii) all code within the main parentheses of an if, switch or range-based-for statement, up to and including the last semicolon, if any, consists of init-statements.